DEPI GRADUATION PROJECT

SYSTEM ANALYSIS & DESIGN

AI: The Global Transformation

AI Impact Intelligence Dashboard

Project information	Details
Track	Data Analysis
Group code	ONL5_DAT1_G2
Team Lead	Omima Mohamed Alam El-Din Ali
Team members	Sayed Ahmed Sayed Farag; Eman Abdullah Abu Bakr; Rana Raafat Abdel Hamid Qasem; Osama Ayman Hosny Abdel Hady
Core dataset selected	Global Artificial Intelligence Indicator Database (GAID), Version 2, 1998–2025
Supporting data	Company AI adoption indicator series; optional Kaggle AI impact dataset after file-level validation
Document status	Revised proposal — final schema and measures depend on downloaded source files and codebook audit

Prepared for DEPI — Data Analysis Track

Document Control and Important Data Note

This version revises the earlier System Analysis & Design proposal to use GAID Version 2 as the primary global indicator source. The Kaggle dataset supplied by the team is not discarded: it may be retained as a supplementary source for industry-level adoption, workforce and revenue-impact analysis if its downloadable file, provenance, definitions, and license pass the validation steps below.

The online GAID description reports a harmonized long-format database covering 1998–2025, 227 countries/territories, 259,546 observations, and 24,453 indicators, with inputs from international sources including Epoch AI, UNESCO, IEA, WIPO, MacroPolo, Coursera, and the World Bank. These are dataset-level claims from its repository description, not results independently recalculated by the project team.

The full GAID file and codebook have not yet been downloaded and inspected in this work session. Therefore, the exact column names, indicator availability by year/country, missingness, and the presence of specific investment or company-performance measures remain subject to a data audit. The logical model below is intentionally proposed, not represented as the verified physical schema.

A country-level competitiveness or AI-readiness score is not the same thing as company-level financial performance. The project will not claim that GAID directly measures firm profitability, productivity, or market-share changes unless a verified source field supports that claim.

1. Executive Summary

AI: The Global Transformation is a data analytics and business-intelligence project that investigates how artificial intelligence is developing across countries and how this development relates to investment, economic capacity, innovation, readiness, and business adoption. The deliverable is an interactive dashboard supported by a reproducible data preparation workflow, source catalog, data dictionary, and documented analytical methods.

The revised design uses GAID as the primary country-year indicator repository. A carefully selected subset of indicators will be extracted from the master long-format file based on the codebook. A secondary company-adoption series may be added to represent organizational AI use. The team-provided Kaggle dataset may be used only for supplementary industry-level analysis after inspecting its actual CSV and provenance.

2. Problem Statement

AI investment, adoption, innovation, and readiness are often reported across different institutions and formats. This makes it difficult for a non-specialist audience to compare countries over time, understand the relationship between AI development and broader economic indicators, and distinguish verified observations from estimates or survey perceptions.

The project addresses this problem by organizing selected indicators into a traceable analytical model and presenting them through a clear dashboard. It is descriptive and exploratory: observed associations will be reported as relationships, not proof that AI caused an economic or company outcome.

3. Project Aim and Objectives

- Track selected AI investment and AI-economy indicators across countries and years.
- Compare national AI development, readiness, infrastructure, talent, and innovation indicators.
- Explore organizational AI adoption and available measures of perceived competitive advantage or business impact.
- Identify patterns, differences, and gaps across countries, sectors, and time where the sources support those comparisons.
- Build an interactive dashboard with transparent definitions, units, source attribution, and uncertainty notes.
- Create a reproducible process for ingestion, cleaning, validation, analysis, and dashboard refresh.

4. Analytical Framework and Scope

Theme	Main analytical questions	Candidate indicator families (subject to codebook audit)	Expected analysis
A. AI Investment & Economic Growth	How has AI-related investment/development changed over time? Which countries lead?	AI Economy & Investment; private AI investment; AI market/economic indicators where present; infrastructure investment proxies	Time series, country ranking, growth rates, comparisons; currency/unit normalization only where documented
B. AI Development & National Competitiveness	How do readiness, infrastructure, talent, research and innovation differ across countries?	AI readiness/index pillars; compute/data-centre capacity; AI talent; AI patents; skills; governance	Country comparison, maps, rankings, correlation analysis, composite-index interpretation
C. Business Adoption & Competitive Position	How widely are organizations adopting AI, and what business outcomes or perceptions are reported?	Company AI adoption rate by country/year; business-function adoption; survey-based perceived competitive edge; optional industry revenue/workforce indicators	Adoption trend, country/sector comparisons, association analysis; clearly label survey perceptions and estimates

Scope boundary: The third theme will be described as organizational adoption and available business-impact evidence. Company-level financial performance will be included only if the selected source contains identifiable company-level observations with definitions and provenance. Otherwise, the dashboard will not present country-level adoption or competitiveness scores as company performance.

5. Dataset Selection and Source Strategy

Source	Role in project	Why selected / limitation	Required validation
GAID Version 2 (Harvard Dataverse; DOI: 10.7910/DVN/PUMGYU)	Primary source for global AI indicator panel	Broad longitudinal coverage and multiple source domains. Very large indicator catalogue; requires codebook-led selection and careful interpretation of composite scores.	Download master CSV and codebook; inspect physical schema, indicator definitions, units, coverage, duplicate keys, missingness, source metadata, license and version.
Our World in Data — Share of companies using AI technology	Candidate supporting series for company/organizational adoption	Provides a country-level share of companies using AI in at least one business function, sourced from McKinsey surveys via AI Index. Survey-based and not a direct financial performance measure.	Download CSV/metadata; record definition, years, survey basis, weighting, coverage and limitations.
Team-provided Kaggle dataset: Global AI Content Impact Analysis 2020–2025	Optional supplementary industry/workforce/business-impact source	Potentially covers adoption, job displacement, revenue growth, human–AI collaboration and consumer trust, but the exact file and provenance have not been independently inspected.	Team must provide downloaded CSV/XLSX; audit row count, fields, source notes, generation method, missingness, duplicates, units and license before inclusion.

Source links: GAID repository: https://doi.org/10.7910/DVN/PUMGYU | GAID methodology: https://gaid.aiinsocietyhub.com/methodology/ | OWID company AI adoption: https://ourworldindata.org/grapher/share-companies-using-artificial-intelligence | Team Kaggle source: https://www.kaggle.com/datasets/mohanz123/global-ai-content-impact-analysis-2020-2025

The GAID project description states that its master file is organized in long format and that a detailed codebook defines the indicator dictionary, units and ranges. The team must use that codebook rather than infer meanings from indicator labels.

6. Data Acquisition and Validation Plan

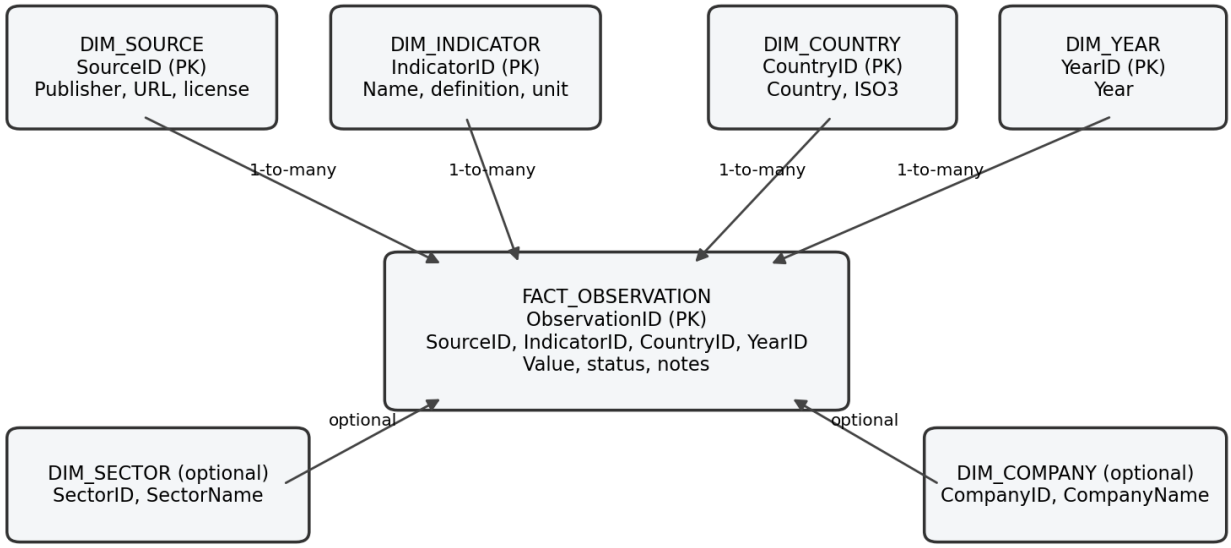
1. Download the exact GAID Version 2 master CSV and its accompanying codebook from the repository; preserve the original files unchanged.
2. Record dataset title, creator, DOI, version, publication date, download date, license, source URL, file name, and checksum in the source catalog.
3. Profile the actual schema: column names, data types, row count, distinct countries, years, indicators, sources, and candidate key uniqueness.
4. Use the codebook to shortlist indicators for each project theme. Record exact metric name, definition, unit, aggregation rule, source, coverage, and known limitations.

5. Filter to a manageable analytical scope (initially 2015–2025 or the longest consistent period available for each chosen indicator); do not force all indicators to share the same period.
6. Check duplicates, invalid country codes, impossible values, mixed units, missing values, and inconsistent source vintages. Do not convert missing observations to zero.
7. Validate selected country-year-indicator records against source metadata or source publications where feasible.
8. Download the OWID supporting series only if needed for Theme C, then harmonize country identifiers and year fields.
9. Inspect the Kaggle file before deciding whether to integrate it. If its provenance or measurement definitions are unclear, exclude it from the analytical model and document the reason.

7. Proposed Logical Data Model

The model below is a star-schema-style analytical design. It is not a claim about the raw GAID column names. The actual ETL mapping must be written after the file and codebook audit.

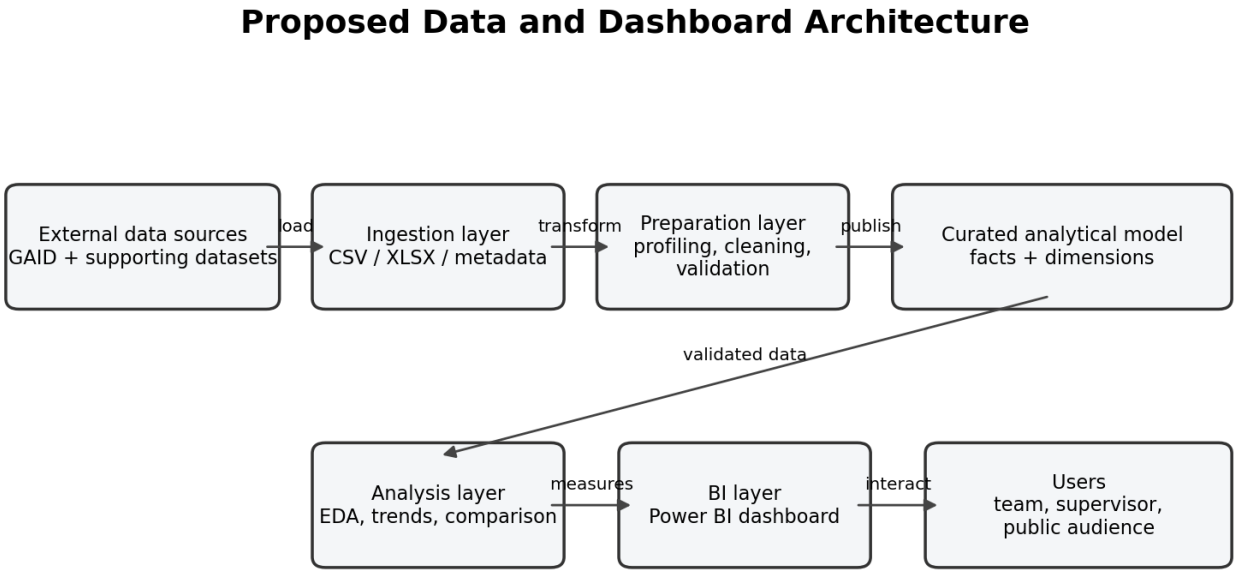
Logical Data Model (to be finalized after schema inspection)



Entity	Purpose	Proposed attributes
FACT_OBSERVATION	One measured indicator value for a defined geography and time point.	ObservationID; SourceID; IndicatorID; CountryID; YearID; Value; observation status; source note
DIM_INDICATOR	Defines what is being measured.	IndicatorID; canonical name; definition; domain/theme; unit; value type; aggregation guidance
DIM_COUNTRY	Standardizes geography.	CountryID; country name; ISO3 code; region/income group if reliably sourced
DIM_YEAR	Supports time analysis.	YearID; calendar year
DIM_SOURCE	Supports traceability.	SourceID; publisher; dataset/report title; URL/DOI; version; license; access date
DIM_SECTOR (optional)	Used only if validated data contains sector dimension.	SectorID; sector name; classification source
DIM_COMPANY (optional)	Used only if a verified company-level source is added.	CompanyID; company name; industry; country; source reference

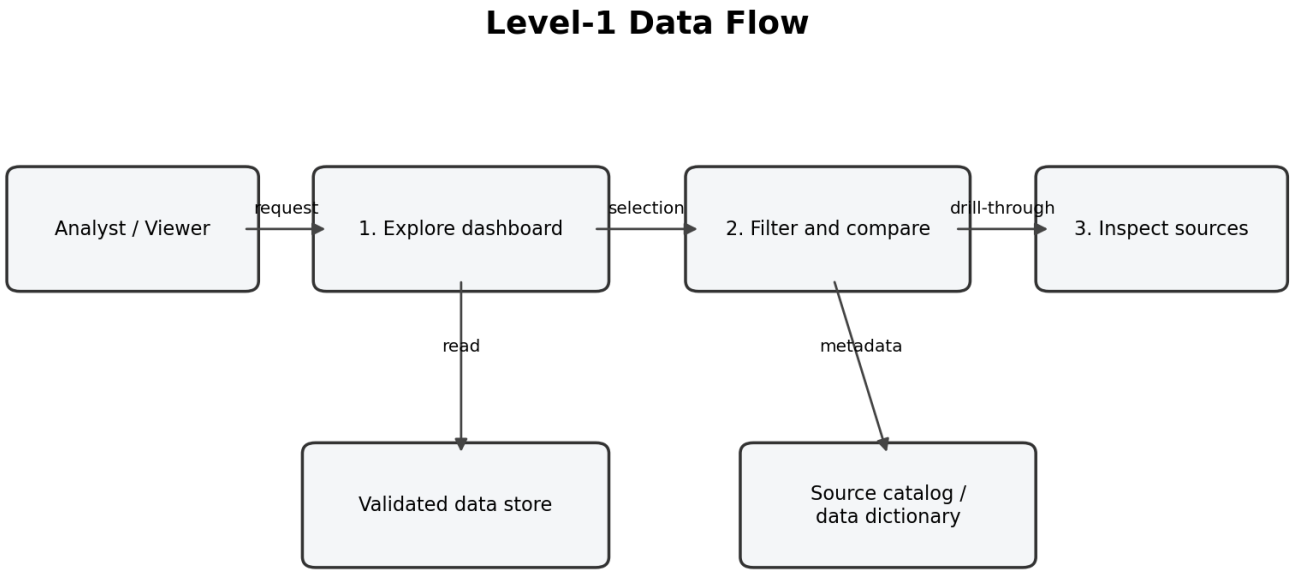
Recommended grain: one row per indicator × country × year × (sector/company only when present in the verified source). If the GAID raw data uses a different grain or multiple observations per combination, retain the original observation identifier and document the aggregation rule. Do not aggregate unlike units or vintages.

8. Proposed System Architecture



Suggested tools: Python (Pandas) for data profiling, cleaning and repeatable transformations; Excel for quick inspection where useful; Power BI for the interactive dashboard; CSV/Parquet for curated intermediate outputs. SQL Server is optional and should be introduced only if the team needs relational storage or the supervisor requires it.

9. Data Flow



10. Source files enter a raw/immutable folder and are registered in the source catalog.
11. Profiling and validation produce a quality report before any data is published.
12. Cleaning and harmonization create curated observation and dimension tables.
13. Analysis measures and dashboard visuals consume only validated curated data.
14. Users can filter results and inspect source details, definitions, units, and caveats.

10. Functional Requirements

ID	Requirement	Acceptance evidence
FR-01	Maintain a source catalog for every dataset and indicator.	Every displayed metric links to publisher, title, URL/DOI, version, license and access date.
FR-02	Preserve raw source files without overwriting.	Raw files are read-only/versioned; transformed outputs are stored separately.
FR-03	Profile and clean selected datasets.	Documented data types, missingness, duplicates, invalid values and transformation steps.
FR-04	Validate the selected indicator subset.	Quality report flags failed checks; failed refresh is not published.
FR-05	Maintain a data dictionary.	Each metric has a definition, unit, source, coverage and interpretation note.
FR-06	Build structured analytical tables.	Fact and dimension tables have documented grain and relationships.
FR-07	Provide reproducible transformations.	A team member can rerun the pipeline and reproduce the curated output.
FR-08	Provide an overview dashboard.	Overview KPIs show label, unit, time period and source context.
FR-09	Provide navigation across three themes.	Users can open Investment, National Development, and Business Adoption pages.
FR-10	Provide filters where supported.	Year, country, indicator, and sector filters update applicable visuals.
FR-11	Show time-series trends.	Charts use actual observation years and visibly distinguish gaps.
FR-12	Compare countries and available sectors.	Comparisons use consistent definitions, units and time coverage.
FR-13	Explain metric definitions.	Tooltips/glossary expose definitions and units.
FR-14	Support source traceability.	A chart can be traced to the underlying source and transformation.
FR-15	Label observations, estimates and survey results.	Visuals clearly label evidence type and uncertainty.
FR-16	Perform descriptive/statistical analysis.	Summary statistics and correlations are reproducible and appropriately caveated.
FR-17	Forecast only when justified.	Forecast includes holdout evaluation and is omitted when series quality is insufficient.
FR-18	Handle unavailable data safely.	Missing categories/years show unavailable or no data, never fabricated zeroes.
FR-19	Support company/sector comparison only when available.	No company/sector visual appears unless the source includes that dimension.
FR-20	Provide a dashboard guide and version record.	User guide, data version and refresh date are visible.

11. Non-Functional Requirements

ID	Quality area	Proposed criterion
NFR-01	Performance	Dashboard pages should load smoothly on the team's target laptop and approved Power BI environment.
NFR-02	Accuracy	Displayed measures must reconcile to curated data and documented calculations.
NFR-03	Integrity	Primary/foreign-key relationships and expected data grain are checked.
NFR-04	Reliability	A failed refresh retains the last validated version and displays its date.
NFR-05	Reproducibility	Code, source versions, transformations and assumptions are documented.
NFR-06	Usability	Navigation, filters and chart labels are understandable to non-technical viewers.
NFR-07	Accessibility	Use readable contrast, descriptive titles, units and non-colour-only cues.
NFR-08	Security & privacy	Do not publish personal or restricted data; respect source terms.
NFR-09	Licensing	Every dataset is used and attributed according to its license/terms.
NFR-10	Maintainability	Indicator mappings and transformation logic are separated and documented.
NFR-11	Portability	Project files run in the agreed team environment with documented dependencies.
NFR-12	Transparency	Forecasts, estimates, survey findings and composite scores are labelled with limitations.

12. User Stories and Use Cases

ID	User story	Acceptance criteria
US-01	As a viewer, I want an overview of AI development so I can understand the main signals quickly.	Each KPI shows name, unit, year, geography and source.
US-02	As an analyst, I want to compare countries and years so I can identify patterns.	Filters apply only where data exists; no-data states are explicit.
US-03	As a reviewer, I want to trace a visual to its source so I can verify it.	Source metadata and transformation notes are available.
US-04	As an analyst, I want to compare investment indicators over time.	Currency, unit, period and nominal/real status are documented.

US-05	As a viewer, I want to understand organizational AI adoption.	Survey definition, population, period and limitations are shown.

UC-01 — Explore the dashboard

Actor: analyst, supervisor or general viewer. Trigger: open dashboard. Main flow: view overview KPIs; choose a theme; inspect trend/comparison; open source notes. Exception: if a source is unavailable or refresh fails, show a clear status and retain the last validated release. Evidence: demonstrate navigation and reconcile one KPI to the curated table.

UC-02 — Filter and compare indicators

Actor: analyst/researcher. Main flow: select theme and metric; choose year range; select country or sector where supported; inspect the updated visual; reset filters. Exception: if no records match, display an empty state rather than zero. Evidence: compare a known filter result with the underlying data.

UC-03 — Trace a metric

Actor: supervisor/reviewer. Main flow: open metric details; review definition, unit, publisher, source URL/DOI, coverage and access date; review transformation notes. Evidence: reproduce one chart value from the source or curated data.

UC-04 — Review a forecast

Actor: analyst/supervisor. Preconditions: adequate historical coverage and a valid holdout evaluation. Main flow: view actual observations separately from estimates; inspect horizon, model, baseline and error metric. Exception: suppress the forecast when the series is too short, inconsistent or poorly evaluated.

13. Analysis Plan and Dashboard Pages

Page	Purpose	Suggested visuals / metrics
1. Executive Overview	Communicate the overall picture and provide navigation.	Selected KPI cards; timeline; country comparison; data coverage note.
2. AI Investment & Economy	Explore investment and AI-economy indicators.	Investment trend; country ranking; growth-rate comparison; unit/currency notes.
3. National AI Development	Compare readiness and enabling capabilities.	Readiness/competitiveness score; talent/skills; compute/infrastructure; patents/innovation; country map.
4. Business Adoption	Show organizational adoption and business-related evidence.	Company AI-use share trend; country comparison; business-function/industry breakdown if supported; survey caveats.
5. Data & Methodology	Make evidence transparent.	Source catalog; metric definitions; data coverage; limitations; refresh date.

Potential analyses include year-over-year change, country ranking, descriptive statistics, correlation matrices, and cluster analysis only if the selected data has sufficient observations and comparable definitions. A regression or predictive model requires a defensible target variable, adequate sample size, leakage checks, and evaluation. Correlation must not be described as causation.

14. Risks, Constraints and Mitigations

Risk	Impact	Mitigation
GAID contains many indicators with different definitions/coverage.	Misleading comparisons and excessive scope.	Use the codebook; shortlist a small, defensible indicator set; document each metric.
Dataset metadata may not support company-level performance.	Theme C may overclaim what is measured.	Rename it organizational adoption/business evidence; add firm-level data only if verified.
Different currencies, units or source years.	Invalid comparisons.	Preserve original units; normalize only with documented conversion rules.
Missing observations or changing survey methods.	Biased trends.	Show coverage and gaps; do not impute without a justified method.
Kaggle dataset provenance is unclear or synthetic.	Weak evidence quality.	Inspect file and methodology; use only as exploratory supplement or exclude.
Forecasts are not reliable for short/inconsistent series.	False confidence.	Use time-aware validation; publish only if performance beats a baseline.
License or attribution restrictions.	Compliance issue.	Review licenses and cite every source; GAID repository lists CC BY-NC 4.0.

15. Testing and Quality Assurance

- Schema test: expected columns and data types are present after mapping.
- Uniqueness test: check the documented observation grain and explain any legitimate repeated records.
- Range test: validate bounded percentages/scores against codebook ranges; do not apply generic limits to unbounded values.
- Completeness test: report missingness by indicator, country and year.
- Consistency test: check country codes, year values, units, source vintages and indicator definitions.
- Reconciliation test: independently recompute selected dashboard KPIs from curated rows.
- Filter test: verify country/year/indicator filters and empty states.
- Traceability test: follow one chart from dashboard to curated row to source metadata.
- Forecast test (if included): time-ordered train/test split, baseline comparison, error metric and residual review.

16. Implementation Phases and Team Work Plan

Phase	Tasks	Deliverable / exit criterion
1. Source audit	Download GAID/codebook; inspect Kaggle file; evaluate OWID series; confirm license and scope.	Approved source inventory and indicator shortlist.
2. Data preparation	Profile, clean, harmonize and validate selected observations.	Reproducible ETL scripts and quality report.
3. Data modeling	Build fact/dimension tables and define metric dictionary.	Documented model and data dictionary.
4. Analysis	EDA, trends, country comparisons, correlation and justified models.	Analysis notebook and documented findings.
5. Dashboard	Build overview and three theme pages with source details.	Interactive Power BI dashboard.
6. QA & presentation	Test measures, filters, traceability, refresh and narrative.	Final dashboard, report, presentation and demo.

Suggested ownership: one member coordinates scope and integration; members own source audit/data preparation, analytical work, dashboard development, and QA/documentation respectively. The team should assign named owners during its planning meeting.

17. Open Decisions Before Physical Database Finalization

- Which exact GAID indicators will represent AI investment, AI economy, readiness, infrastructure, talent and innovation?
- Does the downloaded GAID codebook define the required metrics consistently across the selected period and countries?
- Will the Kaggle file be included after its raw data and source methodology are inspected?
- Will the OWID company-adoption series be included as a supporting source for Theme C?
- Does the supervisor require SQL Server, or are validated CSV/Parquet tables and Power BI sufficient?
- Is the final dashboard expected to cover all countries or a selected comparison group?

18. Conclusion

The revised solution is built around a stronger and more comprehensive global AI indicator source while retaining a validation-first approach. GAID provides a credible starting point for the investment, national development and innovation dimensions; a company-adoption series can support the business theme. The project will only publish indicators that have been verified in the downloaded files and codebook. The final physical schema, calculations and dashboard measures must be confirmed after the source audit.

19. References

- Hung, J. (2026). Global Artificial Intelligence Indicator Database (GAID), 1998–2025 (Version 2). Harvard Dataverse. https://doi.org/10.7910/DVN/PUMGYU
- GAID Project. GAID AI Development Index — Methodology. https://gaid.aiinsocietyhub.com/methodology/
- Our World in Data. Share of companies using artificial intelligence technology. https://ourworldindata.org/grapher/share-companies-using-artificial-intelligence
- Team-provided Kaggle dataset. Global AI Content Impact Analysis 2020–2025. https://www.kaggle.com/datasets/mohanz123/global-ai-content-impact-analysis-2020-2025

Access date for web references: 6 November 2026. Dataset download date and exact file version must be recorded by the team when downloaded.